Struktogramm Editor PWA

Download, Installation & User Guide

Version 2.1 · SVG-rendered Nassi–Shneiderman diagram editor

1 Overview

Struktogramm Editor is a browser-based Progressive Web App (PWA) for drawing Nassi–Shneiderman structured diagrams (Struktogramme). It runs entirely in the browser — no server, no account, no internet connection required after the first load. Diagrams are saved locally as **.stkg** files and can be exported as SVG, PNG, or PDF.

Technology	Plain JavaScript ES2020 modules, SVG rendering, Service Worker PWA
Languages	English and German (switchable at runtime)
Themes	Light, Dark, Eggshell White, Navy Blue Dark, Siemens
File format	.stkg (JSON, human-readable)
Exports	SVG, PNG (2× retina), PDF (browser print dialog)
Tested on	Chrome 120+, Edge 120+

2 Requirements

The following software must be installed before using the app:

Node.js LTS	Version 20 or newer — download from https://nodejs.org
Visual Studio Code	Download from https://code.visualstudio.com
Live Server extension	Install inside VS Code (see Section 3, Step 3)

Important: The app cannot be opened by simply double-clicking index.html. Browsers block Service Workers and certain JavaScript module features when a file is opened directly from disk (file:// protocol). You must use VS Code with the Live Server extension as described below.

3 Installation

Step-by-step setup — follow these steps in order.

1 **Install Node.js**
Go to https://nodejs.org and download the LTS installer (.msi on Windows). Run the installer and leave all options at their defaults. Make sure "Add to PATH" is checked. After installation, open a new Command Prompt and verify: type node --version and npm --version — both should print a version number.

2 **Install Visual Studio Code**
Go to https://code.visualstudio.com and download the installer for your OS. Run the installer with default settings. VS Code is a free code editor that also acts as a local web server for this project.

3 **Install the Live Server extension in VS Code**
Open VS Code. Click the Extensions icon in the left sidebar (or press Ctrl+Shift+X). In the search box type: Live Server Find the extension by Ritwick Dey and click Install. This extension serves the project over http://localhost so the app works correctly.

4 **Extract the project ZIP**
Download the struktogramm-pwa ZIP file. Right-click the ZIP and choose Extract All. Extract it to any folder you like, for example your Desktop.

5 **Open the project folder in VS Code**
Open VS Code. Go to File → Open Folder. Navigate to and select the struktogramm-pwa folder (the one that contains index.html). Click Select Folder. The folder contents appear in the Explorer panel on the left.

6 **Install npm dependencies**
In VS Code, open the integrated terminal: go to Terminal → New Terminal. The terminal opens at the bottom of the screen, already inside the project folder. Type the following command and press Enter:

```
npm install
```

Wait for it to finish. You will see a message like "added 245 packages". This only needs to be done once.

7 **Start the app with Live Server**
In the VS Code Explorer panel on the left, find index.html. Right-click on index.html and choose Open with Live Server. Your default browser opens automatically at http://127.0.0.1:5500 (or a similar localhost address) and the Struktogramm Editor loads.

Setup complete. The app is now running. Every time you want to use it, open VS Code, open the project folder, and right-click index.html → Open with Live Server.

4 Starting the App After First Setup

Once the initial setup is done you do not need to repeat Steps 1–4. To open the app on subsequent uses:

1. Open Visual Studio Code.
2. Go to **File → Open Folder** and select the **struktogramm-pwa** folder.
3. In the Explorer panel, right-click **index.html**.
4. Choose **Open with Live Server**.
5. The app opens in your browser at **http://127.0.0.1:5500**.

Tip: VS Code remembers recently opened folders. Click File → Open Recent to find the project quickly without browsing.

5 Interface Overview

Top bar	File actions: New, Open, Save, Save As, Save SVG, Save PNG, Save PDF
Control strip	Language, theme, Undo/Redo, Duplicate, Delete, Zoom controls, Fit, Validate, Find
Left panel	Block picker (drag blocks onto the canvas) and defined Functions list
Canvas	The diagram editing area — drag, drop, click to edit inline
Breadcrumb	Navigation trail when viewing nested function diagrams
Right panel	Properties (selected block info), Validation warnings, Keyboard shortcuts

Status bar	Autosave status and current selection info

6 Creating a Diagram

6.1 Adding blocks

Drag any block type from the **Block picker** on the left and drop it onto the canvas. A blue insertion line shows where the block will land. Alternatively, hover over any gap between blocks and click the **+** badge.

6.2 Editing block content

Click inside a block to edit its text, condition, or expression directly in the inline input field. Changes are autosaved to browser storage as you type.

6.3 Moving blocks

Drag the block header (the thin grab strip at the top of each block) to move a block to a new position. On touch devices, touch-and-hold then drag.

6.4 Right-click context menu

Right-click any block to access: Open function, Duplicate, Comment, Delete.

6.5 Functions / methods

Add a **Function/Method** block and type the function name. Click the ▼ toggle button on the right side of the block to expand the function body inline. The function also appears in the **Functions** list on the left panel.

6.6 Switch-Case columns

Add a **Switch-Case** block and enter the switch expression. Hover over the divider between case columns to reveal the **+** badge — click it to insert a new case column. Click any case label to rename it.

7 Block Types

Block type	Key	Description
Statement	1	A single instruction or assignment (e.g. x = x + 1)
If / Else	2	Conditional branch — true path left, false path right
While-Do	3	Pre-condition loop — condition checked before each iteration
Do-While	4	Post-condition loop — body executes at least once
For-loop	5	Counted loop with init, condition, and step fields
Function / Method	6	Subroutine call — expandable to show its body inline
Switch-Case	7	Multi-branch selection — one column per case
Break / Return / Continue	8	Flow terminator — select kind from dropdown

Comment / Annotation	9	Non-executable note attached to the diagram

8 Saving and Opening Projects

Save — writes the diagram to the previously saved .stkg file. If no file has been saved yet, a Save As dialog opens.

Save As — always opens a file picker to choose a new save location.

Open — opens an existing .stkg file from disk.

> Note: Save and Save As use the browser's File System Access API, which is supported in Chrome and Edge. If the picker does not appear, the file is automatically downloaded to your Downloads folder as diagram.stkg instead. The diagram is also autosaved to browser localStorage after every change — if you close and reopen the browser your last diagram is restored automatically.

9 Exporting

Save SVG	Downloads a self-contained SVG file. Opens in Inkscape, Illustrator, or any browser. All block geometry and text are preserved as vector graphics.
Save PNG	Rasterises the diagram at 2× resolution and downloads a PNG image. Suitable for inserting into Word, PowerPoint, or reports.
Save PDF	Opens a print-preview window. Use Ctrl+P in the browser and select "Save as PDF" as the printer. Set page orientation to Landscape for wide diagrams.

10 Keyboard Shortcuts

Shortcut	Action
`Ctrl + Z`	Undo last action
`Ctrl + Y`	Redo
`Ctrl + C`	Copy selected block (and all its children)
`Ctrl + V`	Paste copied block at the end of the root diagram
`Ctrl + F`	Open / close the Find-in-diagram search bar
`Delete`	Delete selected block (with confirmation dialog)
`Arrow Up / Down`	Move selection to previous / next sibling block
`Arrow Right`	Move selection into the first child block
`Arrow Left`	Move selection up to the parent block
`Enter / Space`	Focus the inline text input of the selected block
`Escape`	Close context menu or cancel current action
`Ctrl + scroll`	Zoom in / out on the canvas

11 Zoom and Navigation

Zoom + / Zoom – — increase or decrease canvas zoom by 10% per click.

100% — reset zoom to 1:1.

Fit — automatically scales the diagram to fill the visible canvas area.

Ctrl + scroll wheel — zoom in/out with the mouse.

The **breadcrumb bar** above the canvas shows your current position when navigating into expanded function diagrams. Click any crumb to jump back up.

12 Find in Diagram

Press **Ctrl+F** or click the **Find** button in the toolbar to open the search bar above the canvas. Type any text to highlight all matching blocks. Use the ↑ ↓ arrows to jump between matches. Press ✕ or Ctrl+F again to close the search bar.

13 Validation

Click **Validate** in the toolbar to check the diagram for common issues. Results appear in the **Validation** panel on the right. Validation also runs automatically after every change.

The validator checks for:

- Empty statement text
- Missing conditions on If/Else, While-Do, Do-While, For-loop blocks
- Missing function names on Function/Method blocks
- Missing switch expressions on Switch-Case blocks
- Duplicate case labels within a single Switch-Case block
- Duplicate function names across the diagram
- Duplicate function definition names

14 Themes and Language

Use the **Language** dropdown to switch between English and German. All UI labels and validation messages update immediately.

Use the **Theme** dropdown to switch the visual appearance. The selected theme is saved and restored on next open.

Light	Default white and teal — good for printing
Dark	Dark background with cyan accents
Eggshell White	Warm off-white with purple accents
Navy Blue Dark	Deep navy with green and blue accents
Siemens	Siemens brand colours — dark with teal highlights

15 Troubleshooting

App does not load / blank page	Make sure you opened it via Live Server (right-click index.html → Open with Live Server). Do not open index.html by double-clicking it — that uses the file:// protocol which blocks the Service Worker and JavaScript modules.
Live Server not in the right-click menu	The Live Server extension may not be installed. Open VS Code Extensions (Ctrl+Shift+X), search for Live Server by Ritwick Dey, and click Install. Then reload VS Code.

npm install fails	Make sure Node.js is installed and on your PATH. Open a new terminal after installing Node and try again.
Save / Save As does nothing	Chrome and Edge support the file picker. Firefox falls back to a download — check your Downloads folder for diagram.stkg.
Export PNG / SVG appears black	Ensure you are using the latest version of the project files. Old versions of export.js used a CSS-based renderer that does not work in exported files.
Diagram lost after closing browser	The diagram autosaves to localStorage. Private/Incognito mode clears this on close. Use Save to write a .stkg file for permanent storage.
Service worker update toast appears	A new version of the app is available. Click the Reload button in the toast to activate the update.

16 Developer Reference

js/main.js	Bootstrap: loads state, wires store, starts render pipeline
js/state.js	Pure state functions: createBlock, flattenBlocks, normalizeState
js/store.js	Immutable store with undo/redo history and localStorage persistence
js/layout.js	Pure geometry engine — no DOM. Converts block tree to layout records
js/svg-render.js	DOM renderer — converts layout records to live SVG + inline inputs
js/events.js	All DOM event handlers, render pipeline, modal/toast helpers
js/export.js	File I/O: open/save .stkg, export SVG/PNG/PDF
js/validation.js	Project validation — returns array of warning strings
js/i18n.js	Translation dictionaries (EN, DE) and tr() helper
js/constants.js	Shared constants: SVG_NS, H (block heights), BLOCK_LIBRARY

Run these commands from the VS Code terminal inside the project folder:

```
npm test # Run all 140 unit tests
npm run lint # ESLint — 0 errors expected
npm run lint:fix # Auto-fix lint errors
npm run format # Prettier auto-format
npm run format:check # Check formatting
npm run typecheck # TypeScript JSDoc type check
```